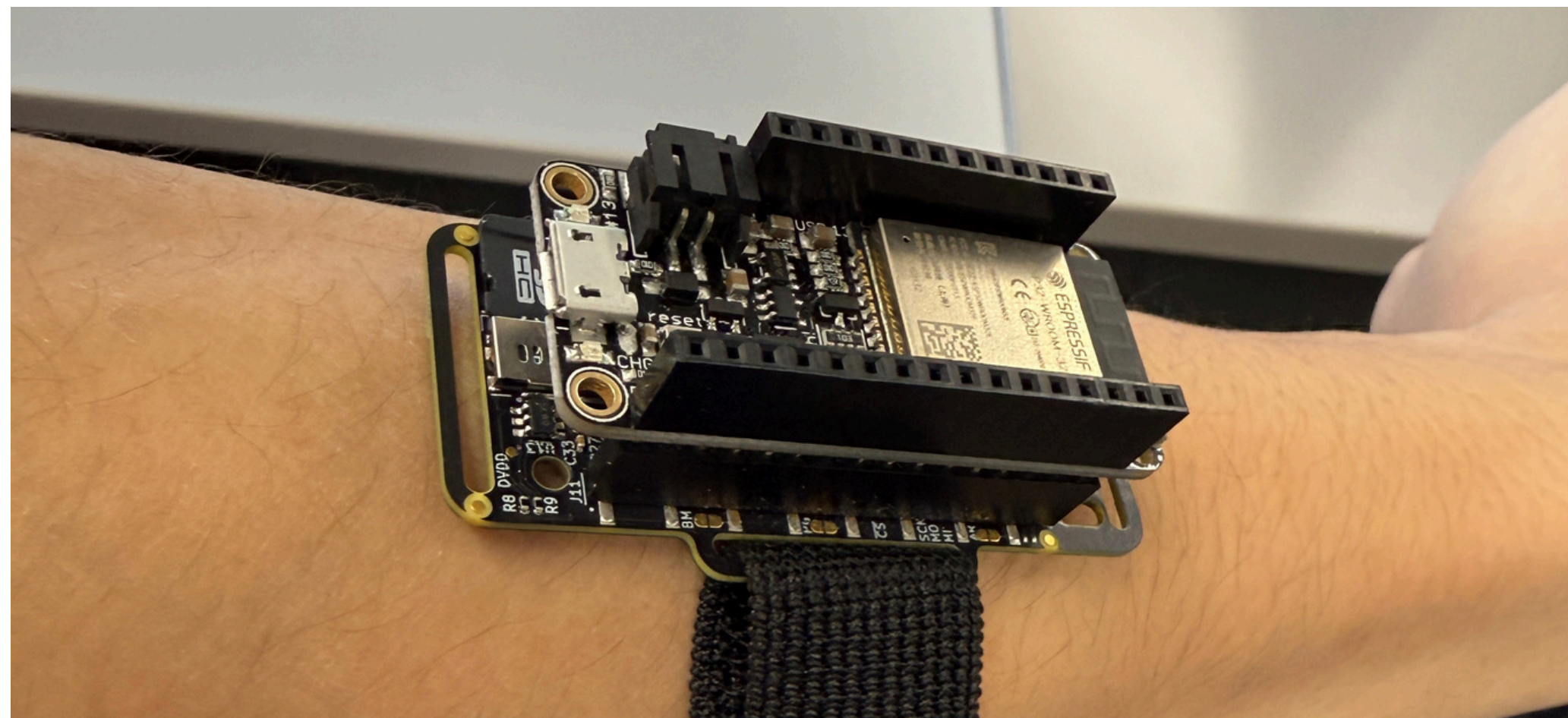


EMOTION BASED MUSIC RECOMMENDATION

Introduction

- Stress and anxiety among adults has risen drastically in the past year.
- Music therapy offers a promising remedy to reduce stress by creating custom playlists that guide a user's mood towards a targeted emotion.
- Current music therapy solutions are clunky and inaccessible, requiring regular consultations with a medical professional.
- AI powered music therapy makes music therapy more flexible and available
- Current AI solutions require taking the user out of the listening experience, requiring the user to perform certain actions during the session.
- Removing these barriers will increase the effectiveness and improve the user experience of AI music therapy.



Objective

- To develop a full stack, wearable integrated emotion aware music recommendation system
- Captures raw ECG and PPG biosignals,
- Predicts the user's current emotional state,
- Generates a gradual playlist transition from current to target emotion.

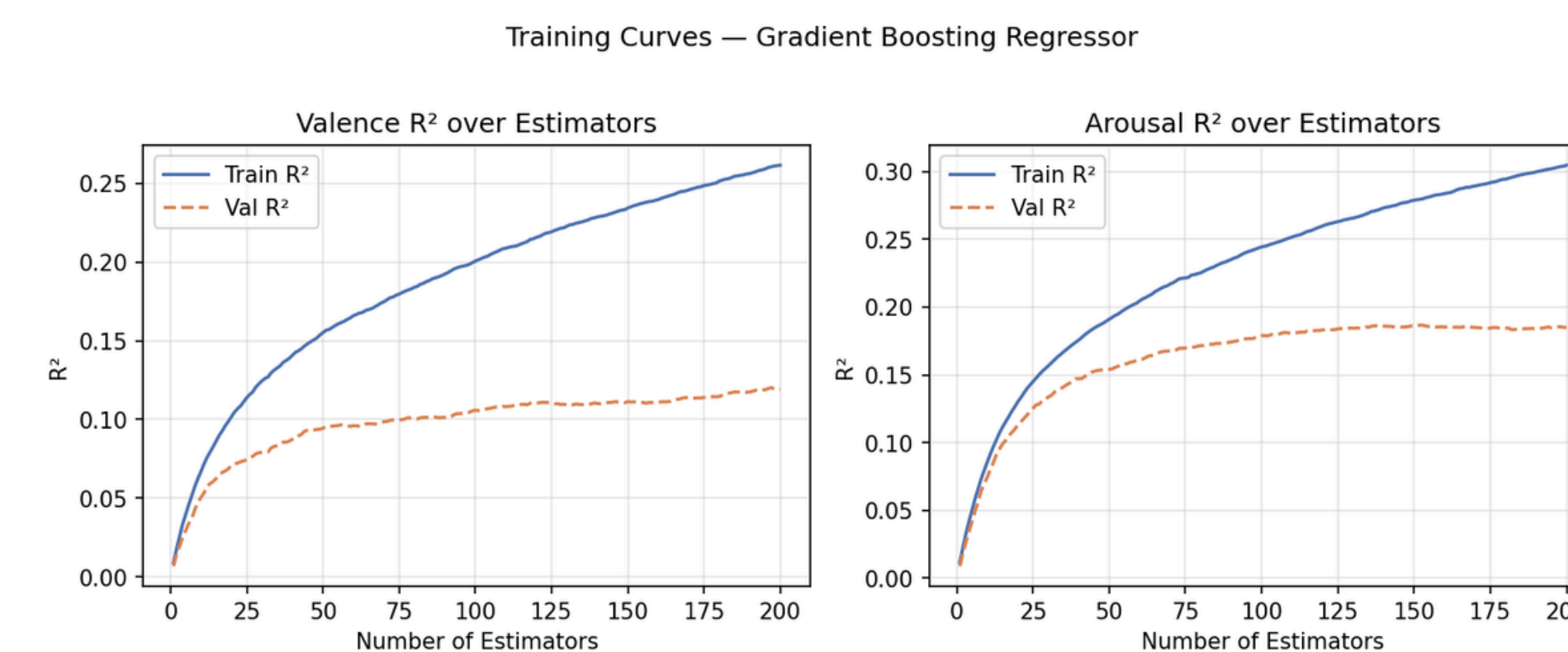


Table 1. R^2 for Valence and Arousal, for training and testing data

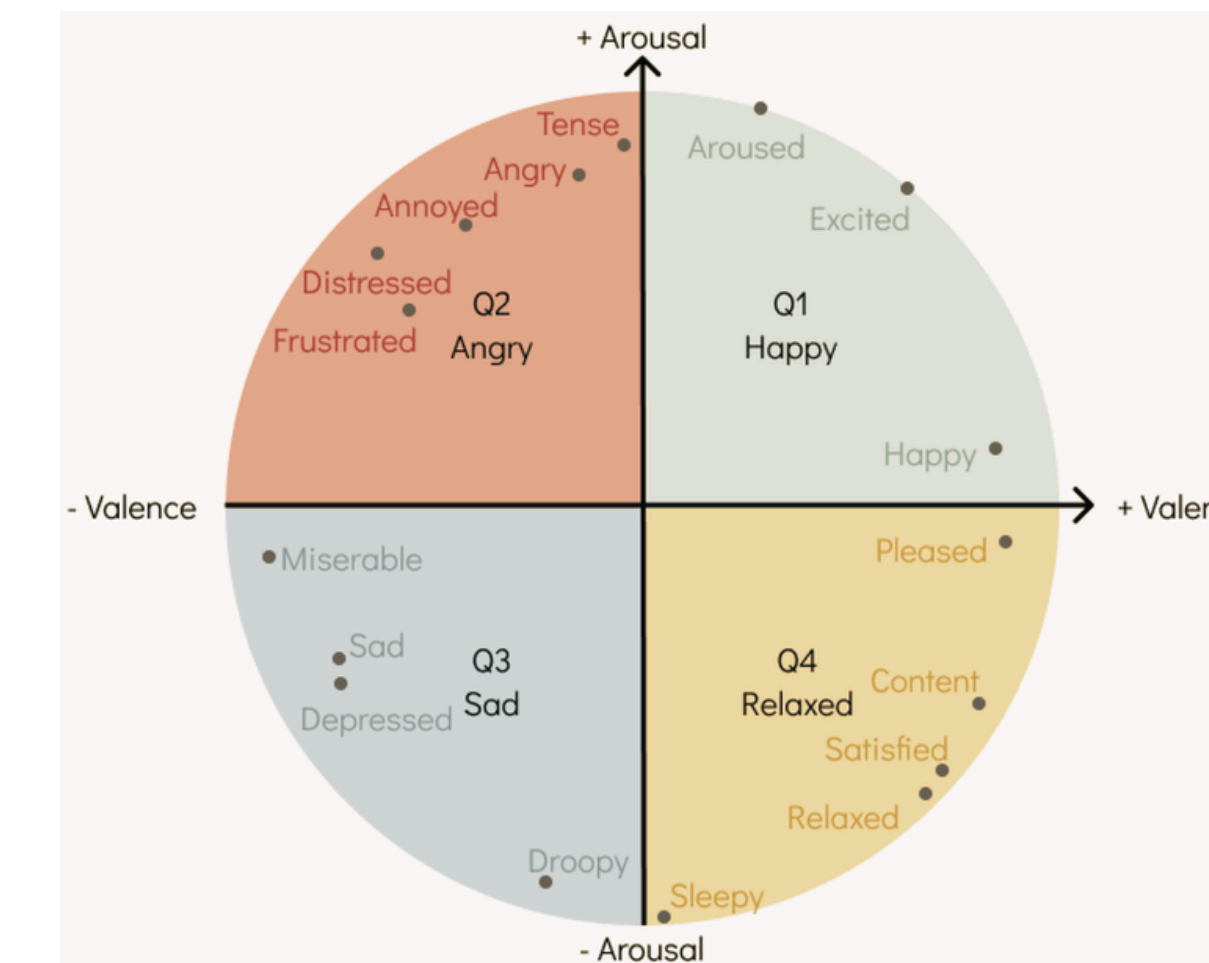


Figure 1. Russell's Circumplex Model of Affect, illustrating how emotions are organized along two continuous dimensions: valence (pleasant-unpleasant) and arousal (activation-deactivation) (Louro et al., 2024).

Methodology

- A wearable device records ECG and EDA signals from the user and exports them as a CSV file, ready for direct upload into the system
- Signals are processed using NeuroKit2. ECG is cleaned and analyzed for HRV metrics: SDNN, pNN50, MeanNN, and EDA is decomposed into phasic and tonic components. Downsized.
- A Random Forest Regressor with MultiOutputRegressor was trained on the CASE dataset using subject specific normalization to account for subject variability
- The predicted valence-arousal coordinates are plotted on Russell's circumplex model of emotion and matched against 90,000 other songs in the MuSe dataset
- A gradual transition playlist is built by interpolating emotional waypoints between the current and target state of emotion, selecting the closest matching song at each step
- The entire pipeline is deployed as a Streamlit web app with two modes: an Upload mode featuring the emotion prediction model and a Demo mode featuring user inputted emotion

Results

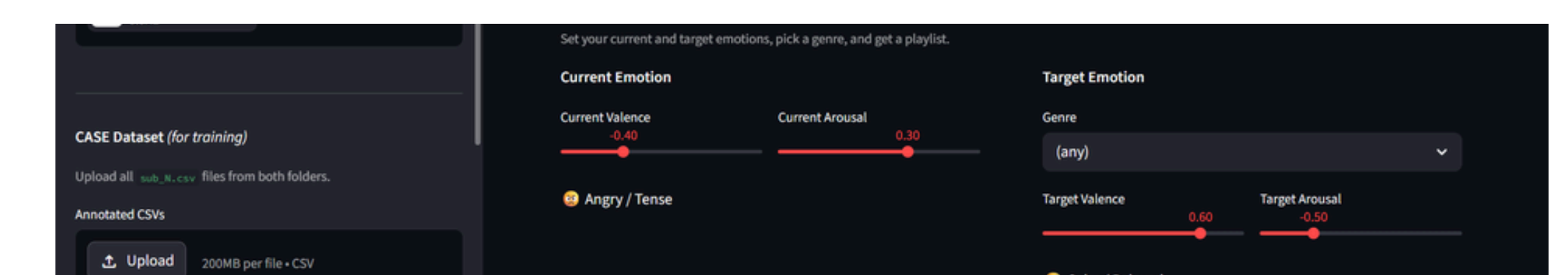
- Fixing a subject specific normalization bug raised R^2 score from -0.08 to 0.119 for Valence and 0.187 for Arousal
- Sensor data is able to be streamed into the model while the previous recommendation is being played
- The recommendation model can reach the targeted emotion within 5 songs

Analysis

The wearable integration reveals that a minimum length of 60-120 seconds is needed for reliable HRV and EDA feature extraction. Physiological emotion responses are highly individual, which is why subject-specific normalization was critical. ECG derived HRV features were significantly more discriminative than EDA alone, though EDA phasic components contributed somewhat to arousal prediction.

Conclusion

The Emotion Aware Music Recommendation project demonstrates a complete closed loop system without requiring any user self-report, if the user so chooses. The wearable captures the data, the pipeline interprets it, and the web app delivers the result. Future work could explore real time streaming directly from the wearable by eliminating the CSV step, deep learning models for better generalization, and music generation..



Sources and Acknowledgements

Liu, C., et al. (2016). Heart rate variability signal features for emotion recognition by using PCA and SVM. In 2016 IEEE Engineering in Medicine and Biology Society (EMBC). <https://doi.org/10.1109/EMBC.2016.7591237>
Russell, J. A. (1980). A circumplex model of affect. Journal of Personality and Social Psychology, 39(6), 1161-1178. <https://doi.org/10.1037/h0077714>
Tsiamartzis, P., et al. (2019). Emotion recognition using electrodermal activity signals and multiscale deep convolution neural network. <https://doi.org/10.1109/TBME.2019.2908892>

Wang, Y., et al. (2023). ECG multi-emotion recognition based on heart rate variability signal features mining. Sensors, 23(20), 8636.
Williams, D., et al. (2020). On the use of AI for generation of functional music to improve mental health. Frontiers in Artificial Intelligence, 3, 497864.